IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science



Data Science Professional
Capstone Project

Brent Gratias
March 29, 2024



Executive Summary

SpaceX provides Falcon 9 rocket launches to clients for millions of dollars less than other providers. Much of these savings are the result of SpaceX ability to reuse the first rocket stage when it lands successfully after launch.

This project will provide a machine learning model that will analyze publicly available data on SpaceX launches to determine the likelihood of a successful first stage landing.

This tool will allow our company to predict SpaceX costs to provide a given launch. This will give our company insights allowing us to successfully bid on these launches.

Introduction

SpaceX provides Falcon 9 rocket launches to clients for a cost of 62 million dollars; other providers cost upward of 165 million dollars. Much of these savings are the result of SpaceX ability to reuse the first rocket stage if it lands successfully after launch.

There is an abundance of publicly available data on SpaceX launches. We wish to explore the correlation between these mission parameters and determine if they have any impact on the likelihood of a successful first stage landing.

Once the most impactful parameters have been identified, a machine learning model will be built, and trained on these specific parameters.

Finally, we will test this tool to determine if it will be effective at predicting the success or failure of subsequent launches.

Section 1

Methodology



Summary of Methodologies

Data collection:

- Retrieved launch data using calls to the Open-Source SpaceX REST API
- Web scraped data from Wikipedia's List of Falcon 9 and Falcon Heavy launches

Data Wrangling:

- Remove missing values
- Programmatically categorize successful and unsuccessful launches

Perform exploratory data analysis (EDA) using visualization and SQL

Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using Scikit-Learn machine language classification models

- Standardize the data, Build, Train and Tune classification models
- Evaluate the accuracy of each model

Data Collection

Data was collected from two sources for our analysis:

API requests from the open-source SpaceX REST API for for launch, rocket, core, capsule, starlink, launchpad, and landing pad data.

Flight Number
Date
Booster Version
Payload Mass
Orbit
Launch Site
Outcome
Flights

Grid Fins
Reused
Legs
Landing Pad
Block
Reused Count
Serial No.
Latitude / Longitude

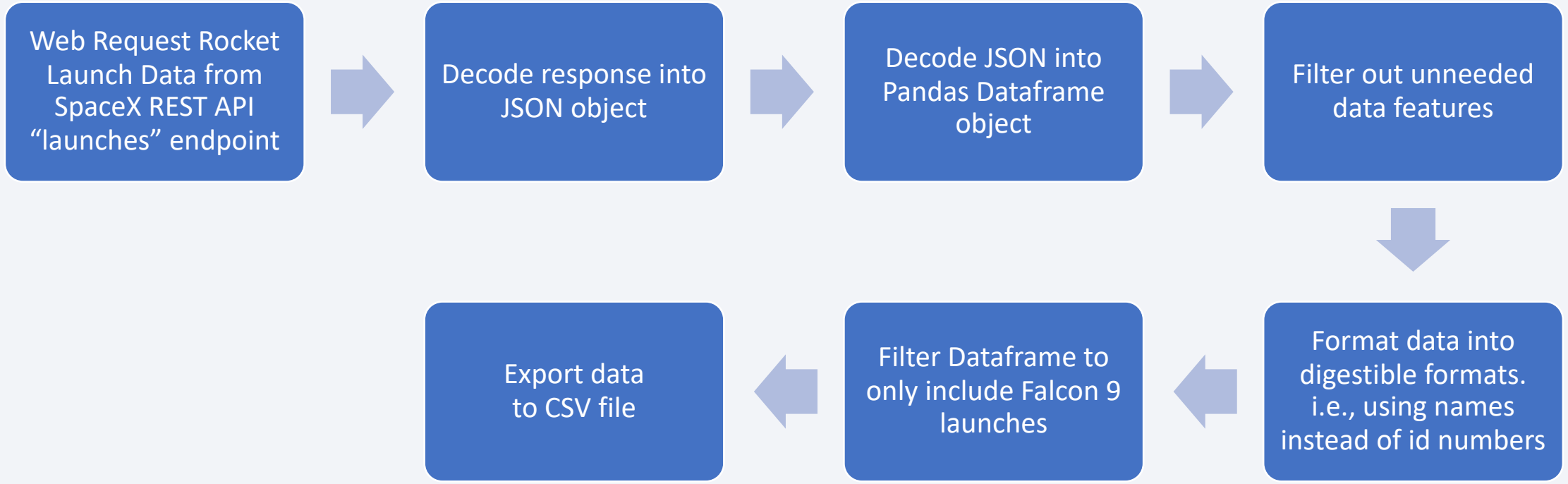
Web scraping data from from **Wikipedia's** List of Falcon 9 and Falcon Heavy launches.

Flight Number
Launch Site
Payload
Payload Mass
Orbit
Customer
Outcome
Booster Version

Booster Landing
Date
Time

Two data sources were required in order to obtain complete information on each launch. These two sources were combined into a single dataset for our analysis.

Data Collection – SpaceX REST API



Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/01-jupyter-labs-spacex-api.ipynb

Data Collection – Web Scraping

Web Request HTML "List of Falcon Launches" webpage from Wikipedia

Use BeautifulSoup library to create BeautifulSoup object from the HTML

Extract the Launches Table from the rest of the webpage

Iterated thru the table tags to extract tabular data into a Dictionary

Pass the dictionary into a Pandas DataFrame

Export the data to a CSV file



2020 [edit]

In late 2019, Gwynne Shotwell stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,[490] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's Long March rocket family.[491]

[hide] Flight No.	Date and time (UTC)	Version, Booster[a]	Launch site	Payload[c]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21[492]	F9 B5 △ B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)
	Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations.[493]								
79	19 January 2020, 15:30[494]	F9 B5 △ B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test[495] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital[496]	NASA (CTS)[497]	Success	No attempt
	An atmospheric test of the Dragon 2 abort system after Max Q. The capsule fired its SuperDraco engines, reached an apogee of 40 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously stated to be accomplished with the Crew Dragon Demo-1 capsule,[498] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019.[418] The abort test used the capsule originally intended for the first crewed flight.[499] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted.[500] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.								
80	29 January 2020, 14:07[501]	F9 B5 △ B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)
	Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the fairing halves was caught, while the other was fished out of the ocean.[502]								
81	17 February 2020, 15:05[503]	F9 B5 △ B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Failure (drone ship)
	Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship[504] due to incorrect wind data.[505] This was the first time a flight proven booster failed to land.								
82	7 March 2020, 04:50[506]	F9 B5 △ B1059.2	CCAFS, SLC-40	SpaceX CRS-20 (Dragon C112.3 △)	1,977 kg (4,359 lb)[507]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
	Last launch of phase 1 of the CRS contract. Carries *Bartolomeo*, an ESA platform for hosting external payloads onto ISS.[508] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part.[509] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.								
83	18 March 2020, 12:16[510]	F9 B5 △ B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Failure (drone ship)
	Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019).[511] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Merlin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit.[512] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor.[513]								
84	22 April 2020, 19:30[514]	F9 B5 △ B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb)[5]	LEO	SpaceX	Success	Success (drone ship)

Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/02-jupyter-labs-webscraping.ipynb

Data Wrangling

Remove missing values

Missing values were discovered in the PayloadMass data feature

1. Calculate a mean value for the Payload Mass of all launches
2. Insert mean value for all records missing PayloadMass data

Programmatically categorize successful and unsuccessful launches

The Outcome column identifies 8 different types of mission outcomes.

1. Create a list of outcomes we identify as failures.
2. Add a binary classification column to represent the outcome of each launch
3. Iterate over the dataframe comparing each outcome to the list of failure outcomes and categorize each launch as either a success or failure

Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/03-jupyter-labs-data-wrangling.ipynb

Exploratory Data Analysis with SQL

Structured Query Language queries performed on this dataset:

1. List unique launch sites
2. List all launch sites beginning with the characters 'CCA'
3. Calculate total payload mass carried by boosters launched by NASA (CRS)
4. Calculate the average payload mass carried by booster version F9 v1.1
5. Return the date for the first successful landing outcome on a ground pad
6. List all boosters which have successfully landed on drone ships with payload from 4000 to 6000 kg
7. Calculate the total number of successful and failed mission outcomes
8. Return the booster version which has carried the maximum payload mass
9. List all failed landings on drone ships, with booster versions, and launch sites for 2015
10. List all landing outcomes with their count between the dates 2010-06-04 and 2017-03-20

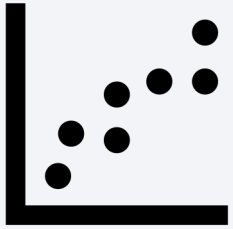
Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/04-jupyter-labs-eda-sql.ipynb

Exploratory Data Analysis with Data Visualization

Visualizations performed on this dataset while exploring insights:

Scatter Charts

- Flight Number vs Launch Site
- Payload vs Launch Site
- Orbit vs Flight Number
- Payload vs Orbit



Bar Charts

- Orbit Success Rates



Line Charts

- Successful launches expressed over time



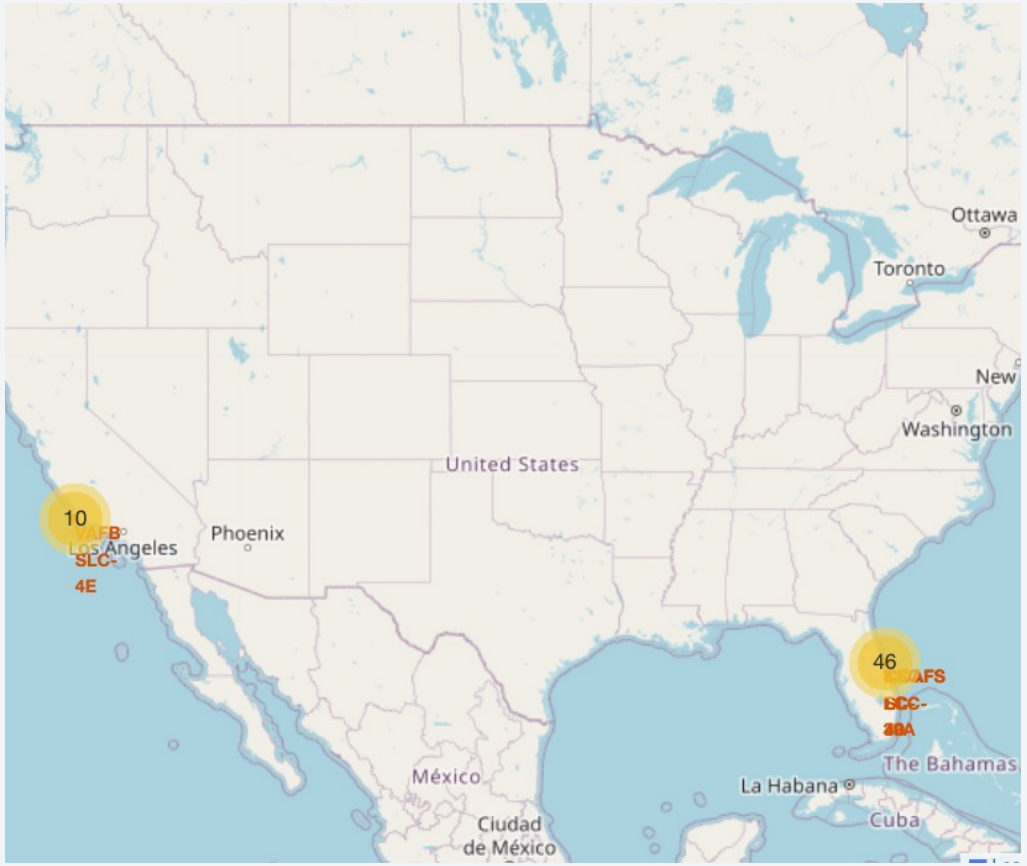
Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/05-jupyter-labs-eda-dataviz.ipynb

EDA Using Interactive Maps with Folium

Folium is a Python library for creating interactive maps. As part of exploratory data analysis, Folium was used to visualize geospatial aspects of each launch site to look for correlation with launch outcomes.

Using Folium:

1. Added markers for each launch site on a world map
2. Colour coded markers to correspond with launch outcomes
3. Calculated distances to proximities like railways, coastlines and nearest cities



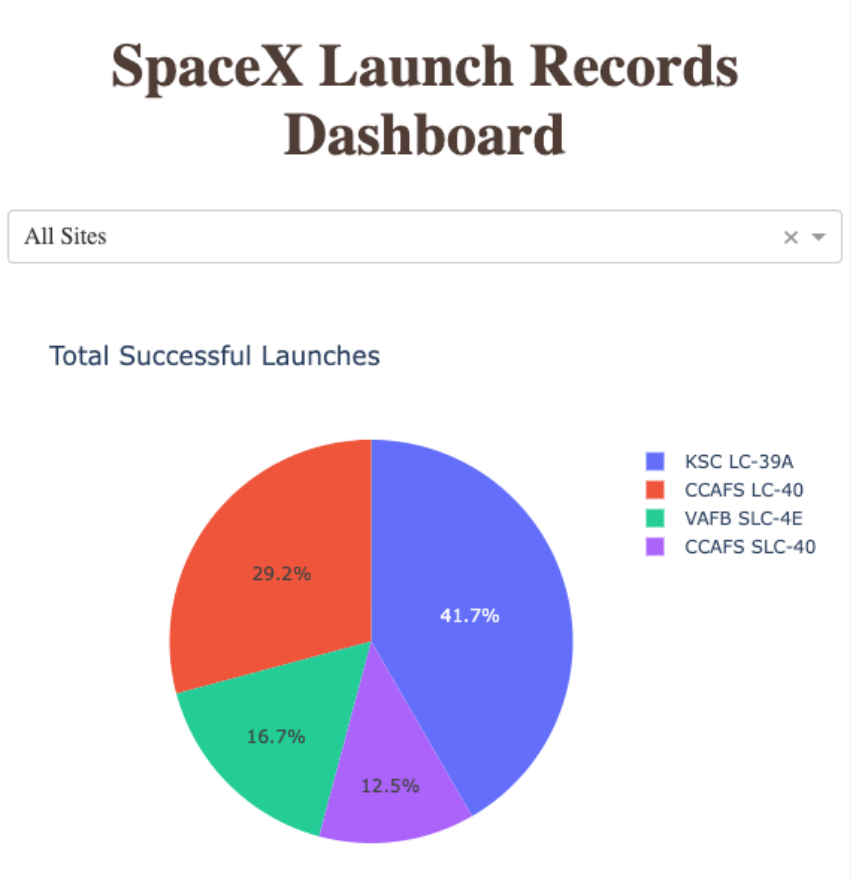
Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/06-jupyter-labs-eda-folium.ipynb

Build a Dashboard with Plotly-Dash

Interactive visualizations on a dashboard allow real-time interactions with visualizations to discover patterns faster and more effectively than static graphs.

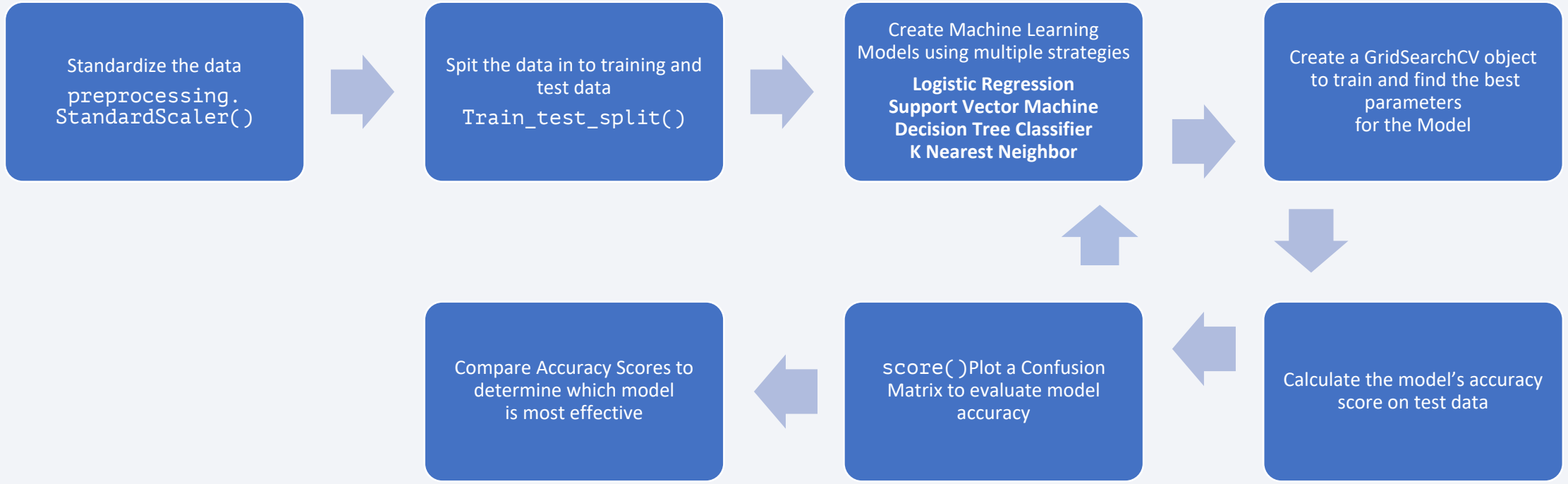
Using Plotly-Dash, an interactive dashboard web application was used to:

1. examine Launch Outcomes by site
2. Examine correlation between payload size and Launch Outcome



Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/07-spacex_dash_app.py

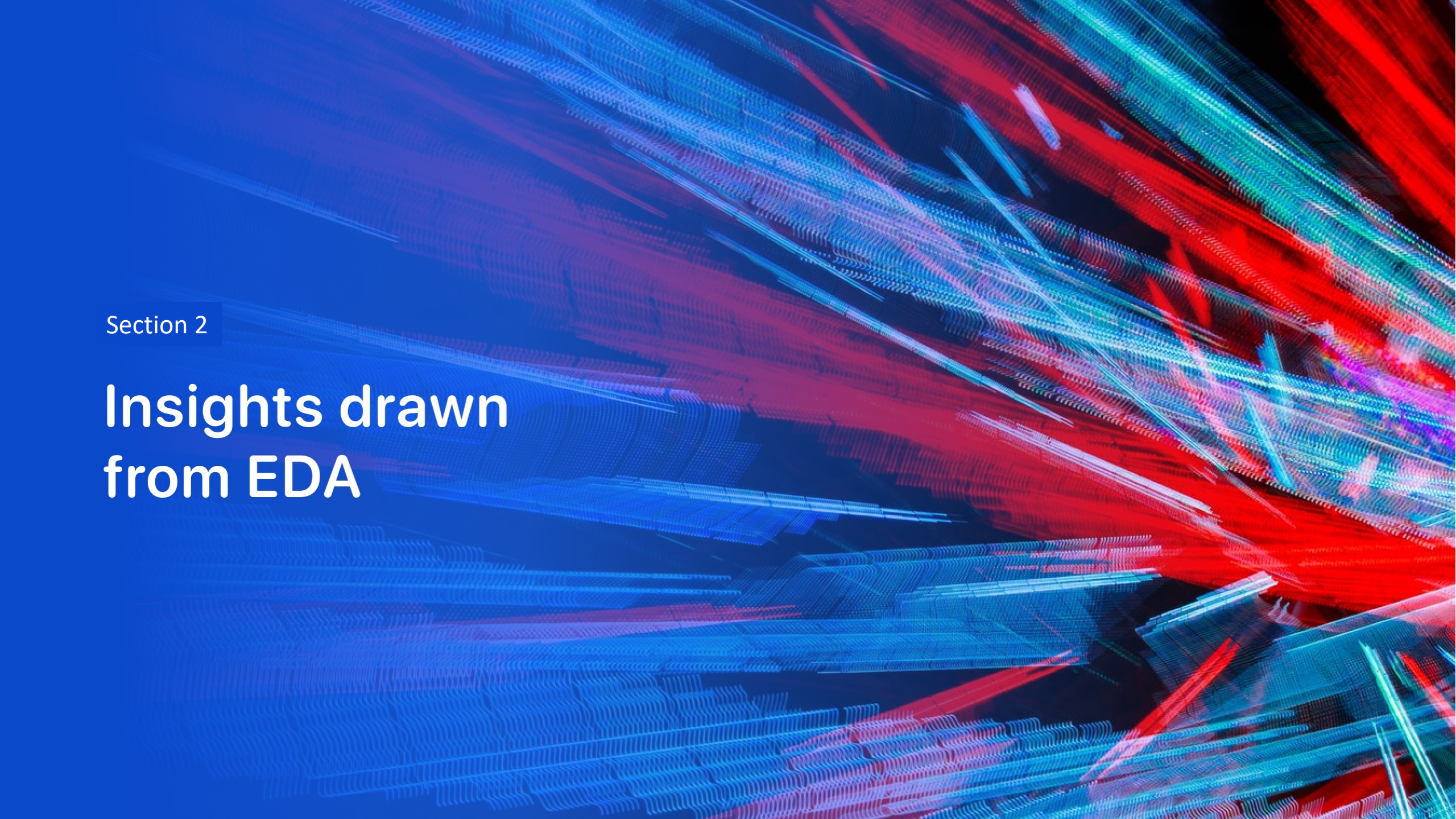
Data Collection – SpaceX REST API



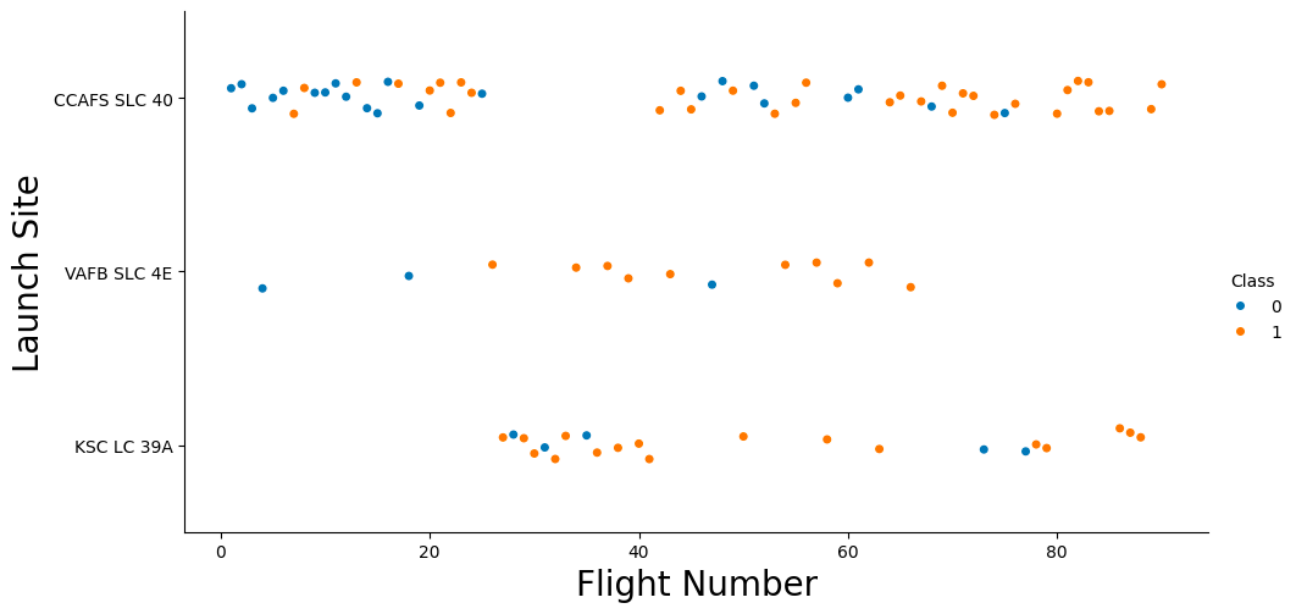
Jupyter Notebook: https://github.com/bgratias/IBM-ADS-Capstone/blob/main/08-jupyter-labs-Machine-Learning-Prediction.ipynb

Section 2

Insights drawn from EDA



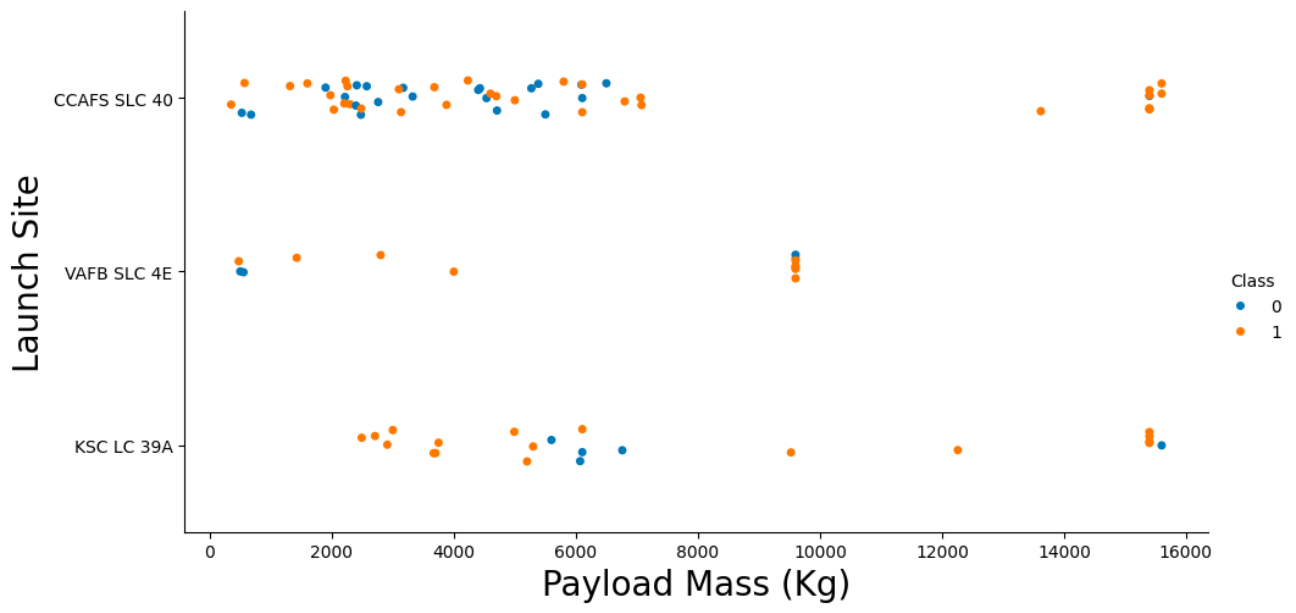
Flight Number vs. Launch Site



Earlier flights were evidently more prone to failure. It can be assumed that each new launch will have a higher chance of success.

Site VAFB SLC 4E and KSC LC 39A have higher success rates overall.

Payload Mass vs. Launch Site

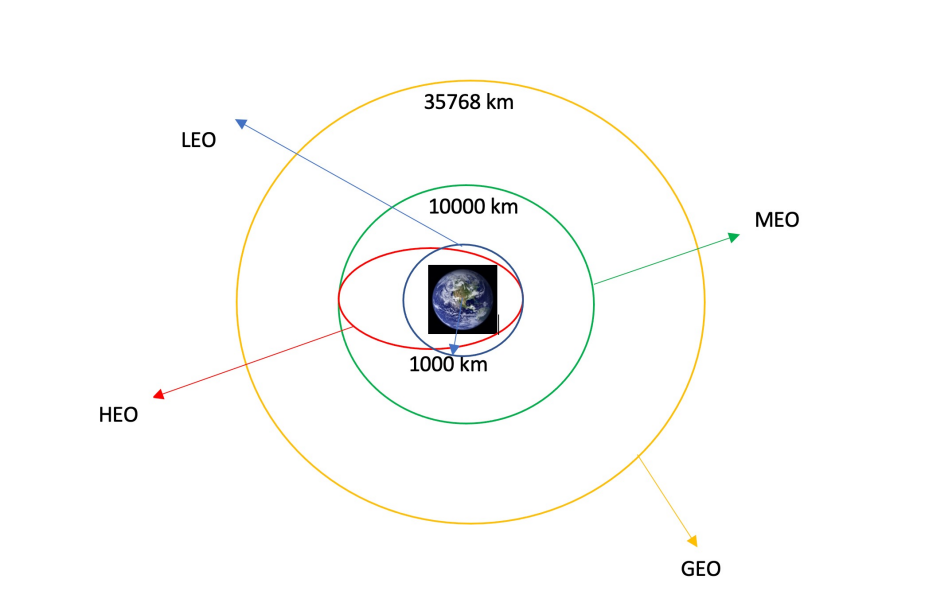


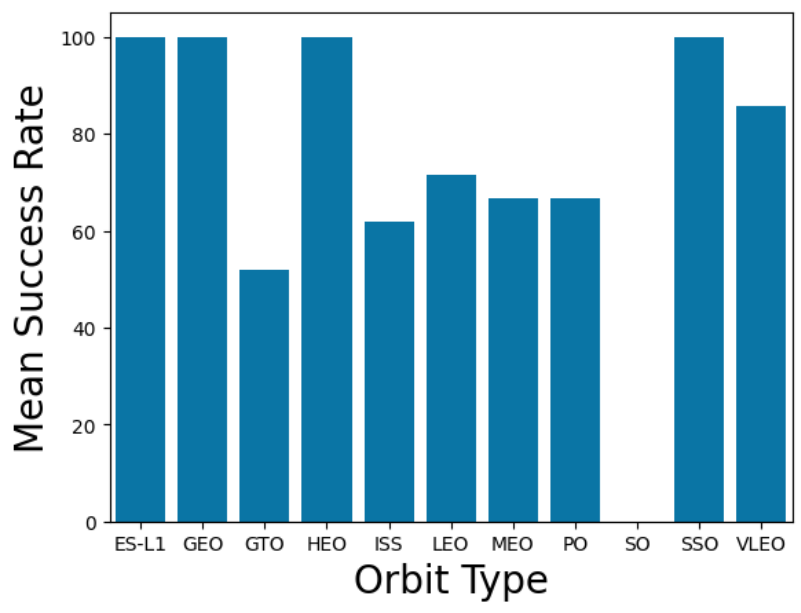
All launch sites appear to relate higher payload mass with higher rates of success.

Almost all launches with a payload mass over 7000 Kg have been successful.

Launch site KSC LC 39A appears to be successful with payloads under 5500 Kg.

Success Outcomes vs. Target Orbit



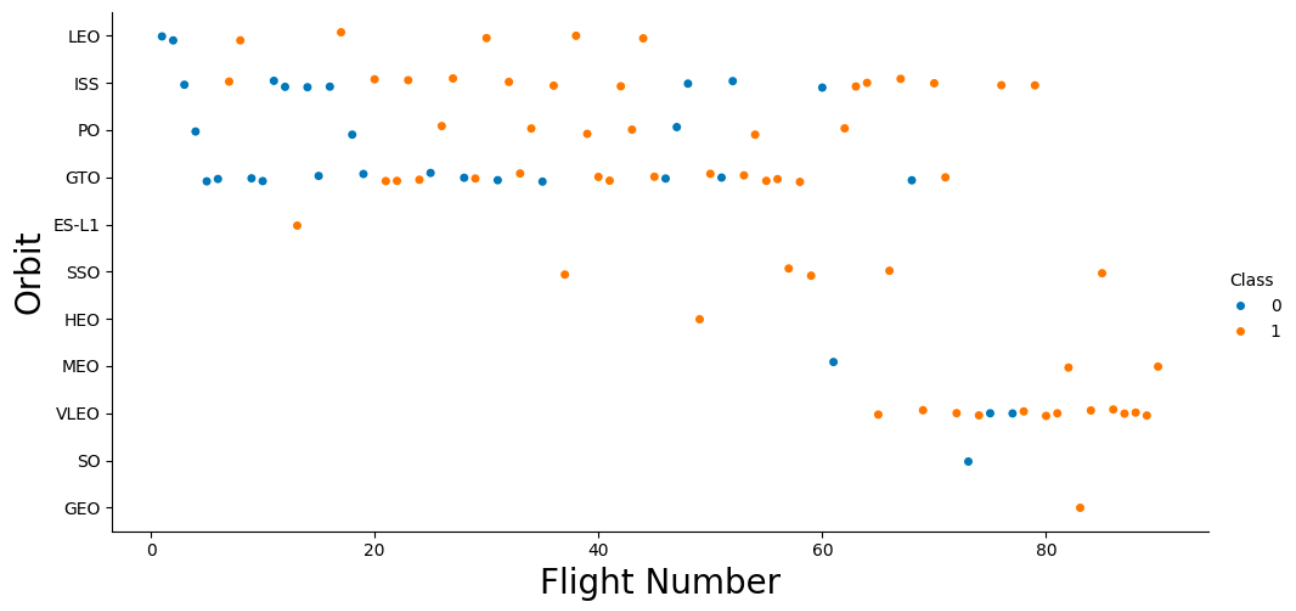


100% Success: Orbits ES-L1, GEO, HEO, SSO

0% Success: Orbits SO

50% - 85% Success: Orbits GTO, ISS, LEO, MEO, PO

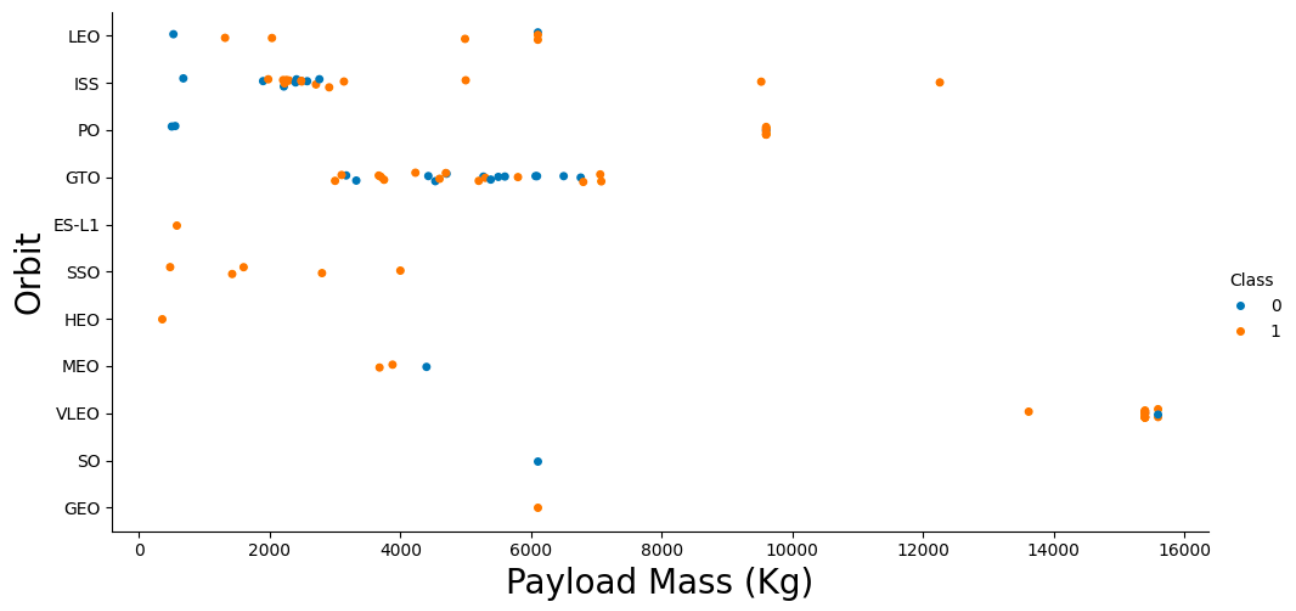
Flight Number vs. Target Orbit



The success of most launches has improved for most target orbits.

GTO, Geosynchronous orbit does not appear to show the same improved success rates over time.

Payload Mass vs. Target Orbit

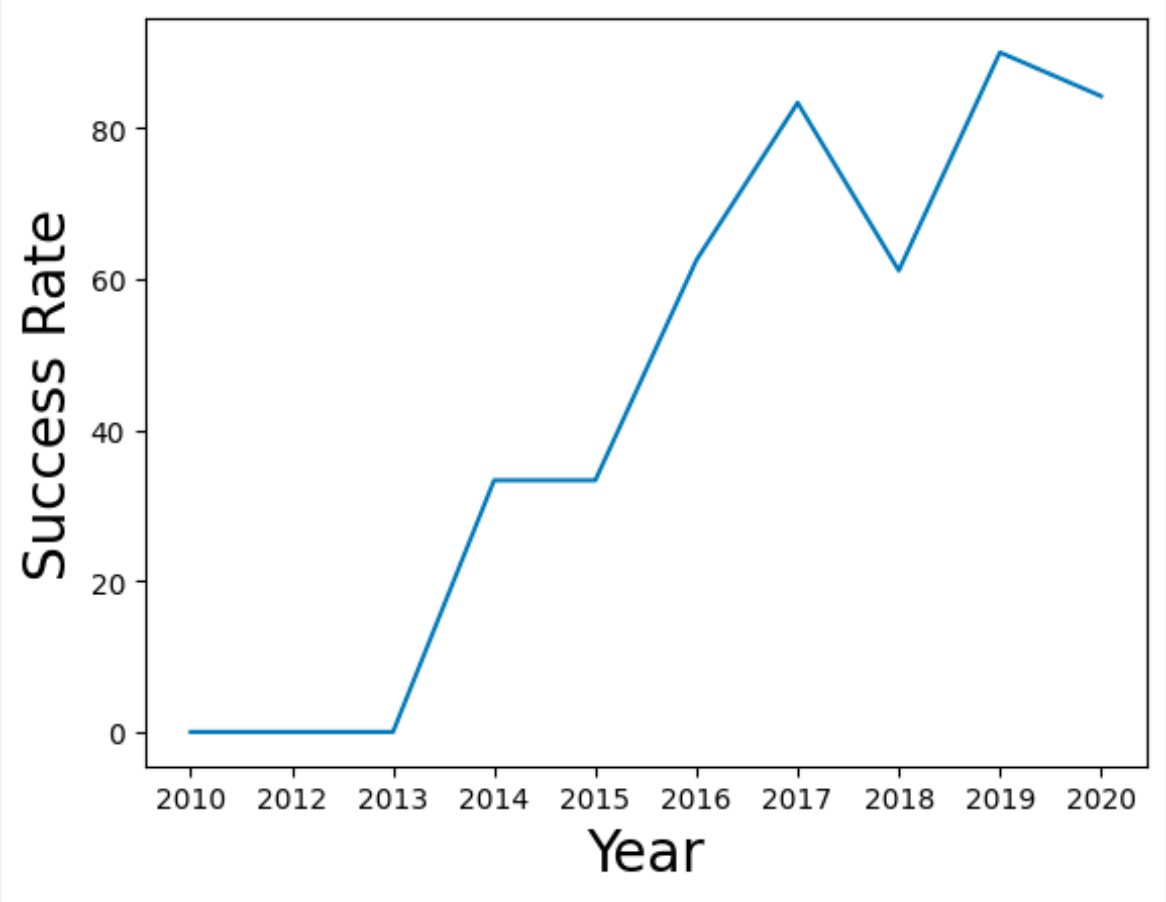


With heavy payloads the successful landing rates are higher for Polar, LEO and ISS orbits.

However, for GTO Geosynchronous orbit targets there appears to be no correlation.

Success Outcomes vs. Target Orbit

Year-on-year, there is clearly a trend toward launches being more and more successful.



EDA using SQL

All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM spacextable WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

EDA using SQL

Total Payload Mass carried by boosters from NASA

```
%%sql SELECT SUM("PAYLOAD_MASS__KG_")
FROM spacextable
WHERE "Customer" = "NASA (CRS)"
```

SUM("PAYLOAD_MASS__KG_")
45596

First successful landing on a ground pad was achieved

```
%%sql SELECT MIN("Date"), "LANDING_OUTCOME"
FROM spacextable
WHERE "LANDING_OUTCOME" = "Success (ground pad)";
```

MIN("Date")	Landing_Outcome
2015-12-22	Success (ground pad)

Average Payload Mass carried by booster F9 ver 1.1

```
%%sql SELECT AVG("PAYLOAD_MASS__KG_")
FROM spacextable
WHERE "Booster_Version" = "F9 v1.1";
```

AVG("PAYLOAD_MASS__KG_")
2928.4

Boosters which have successfully landed on a drone ship with a payload mass between 4000 and 6000

```
%%sql SELECT DISTINCT "Booster_Version", "PAYLOAD_MASS__KG_", "Landing_Outcome"
FROM spacextable
WHERE "Landing_Outcome" = "Success (drone ship)" AND "PAYLOAD_MASS__KG_"
BETWEEN 4000 AND 6000
ORDER BY "Booster_Version";
```

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)

EDA using SQL

Total Number of Success and Failure Mission Outcomes

```
%%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") AS TOTAL
FROM spacextable
GROUP BY "Mission_Outcome";
```

Mission_Outcome	TOTAL
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Booster versions which have carried the maximum payload mass

```
%%sql SELECT DISTINCT "Booster_Version", "PAYLOAD_MASS__KG_"
FROM spacextable
WHERE "PAYLOAD_MASS__KG_" = (
    SELECT MAX("PAYLOAD_MASS__KG_")
    FROM spacextable)
ORDER BY "Booster_Version";
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

EDA using SQL

List of Failed Landing Outcomes for 2015

```
%%sql SELECT SUBSTR("Date",6,2)
AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM spacextable
WHERE SUBSTR("Date",1,4) = "2015" AND "Landing_Outcome" = "Failure (drone ship)"
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") AS "TOTAL"
FROM spacextable
WHERE "Date" BETWEEN "2010-06-04" AND "2017-03-20"
GROUP BY "Landing_Outcome"
ORDER BY "TOTAL" DESC;
```

Landing_Outcome	TOTAL
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1



Section 3

Launch Sites Proximities Analysis

Launch Site Locations

All Launch sites are located on the continental United States.

Launch sites are located in the southernmost part of the country

One launch site is on the West coast and three launch sites are concentrated in very close proximity to each other on the East coast.

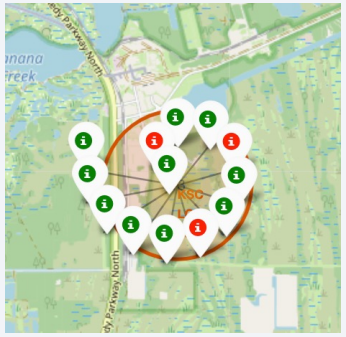


Launch Sites and Launch Outcomes

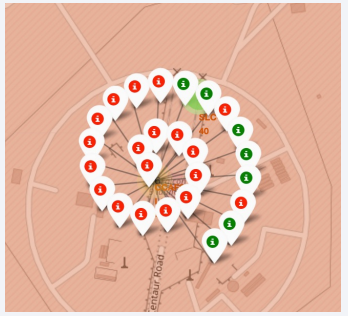
VAFB SLC-4E



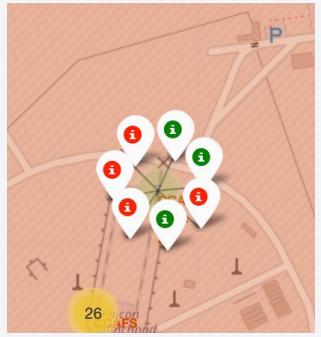
KSC LC-39A



CCAFS LC-40



CCAFS SLC-40



All launch sites have had both successful and failed launches.
The Kennedy Space Centre LC-39A has has the most successful launches and the least failures.

VAFB SLC-4E Launch Site Proximities

Nearest Coastline
1.36 km

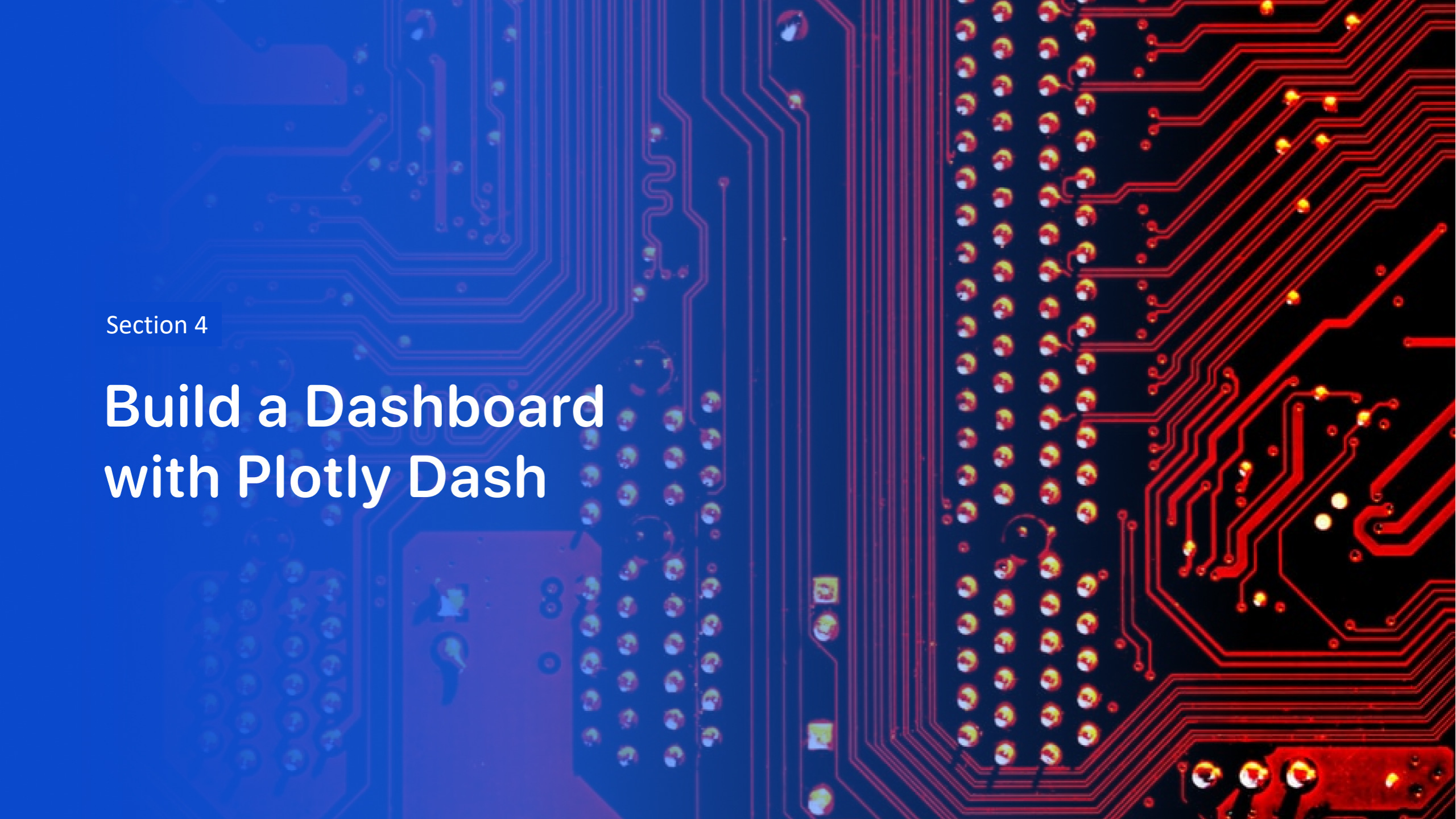
Nearest Railway
1.26 km





Nearest City
39.04 km

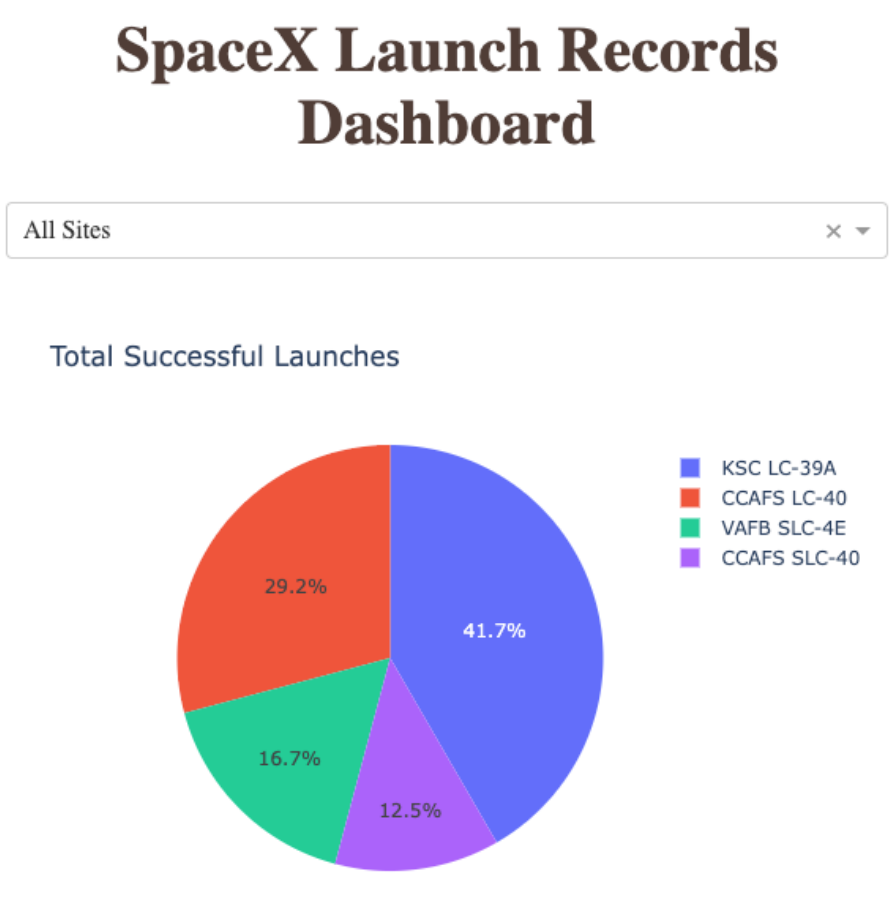
Nearest Highway
14.07 km



Section 4

Build a Dashboard with Plotly Dash

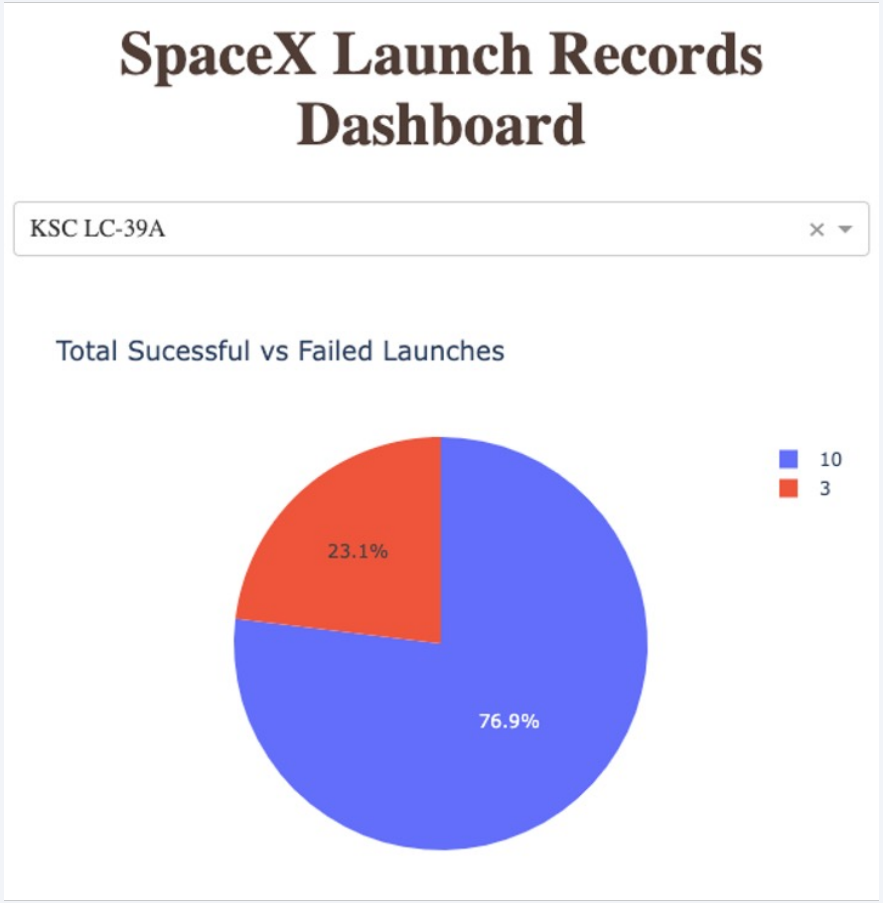
Number of Successful Launches by Launch site



The majority of successful launches have been from KSC LC-39A.

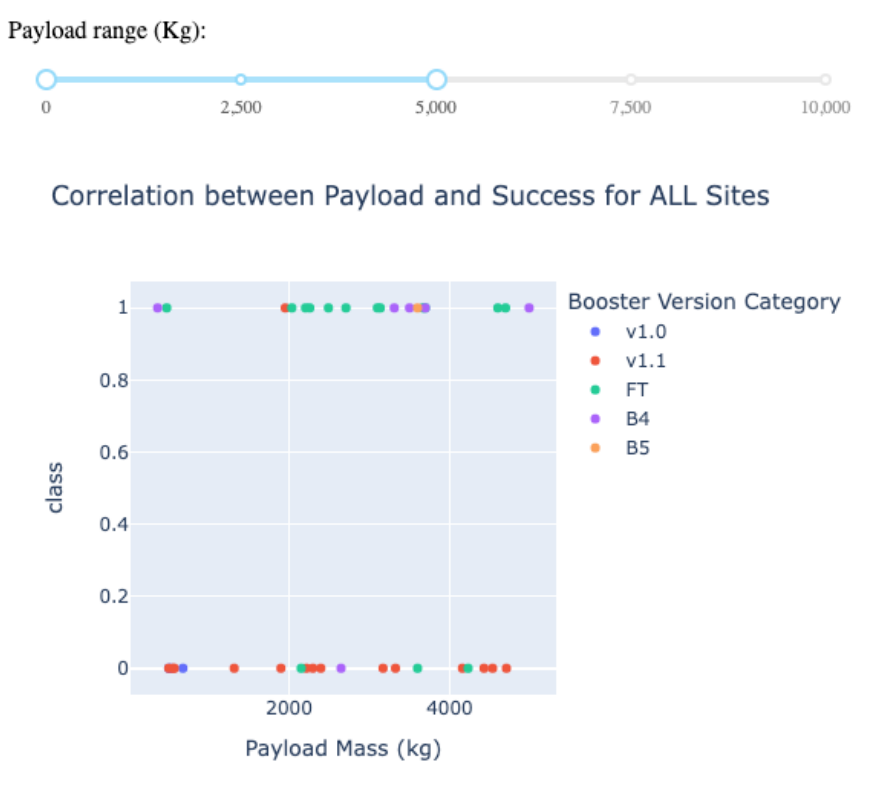
At CCAFS, LC-40 accounts for more than two times the number of successful launches of SLC-40.

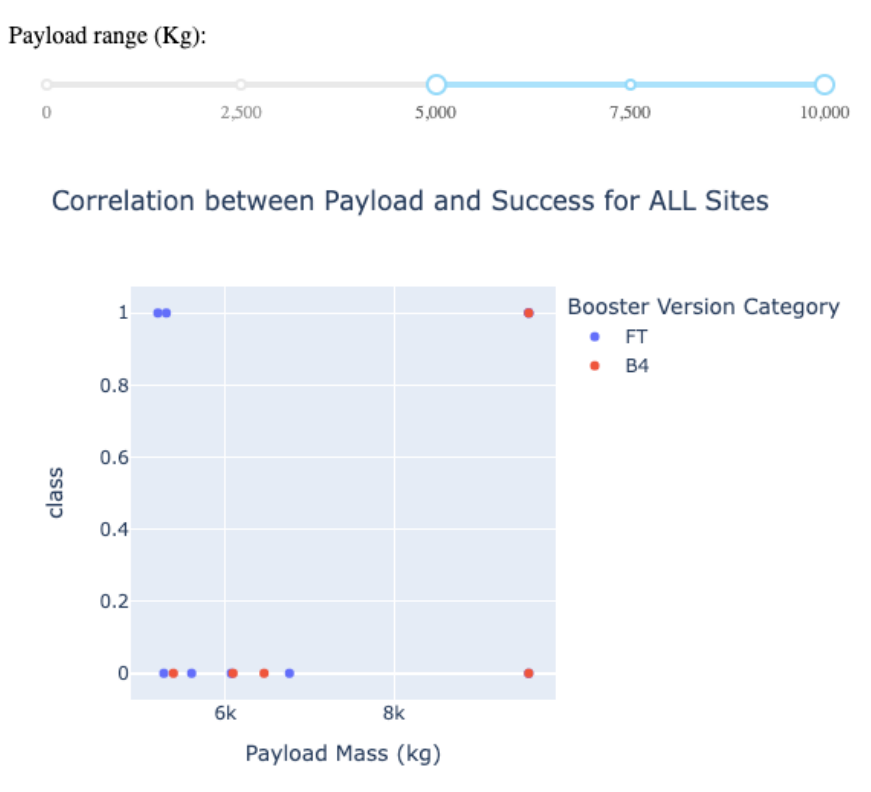
Launch Success Ratio at KSC LC39A



Successful launches far outnumber unsuccessful launches at the Kennedy Space Center LC39A

Payload Mass vs Launch Success for All Launch Sites





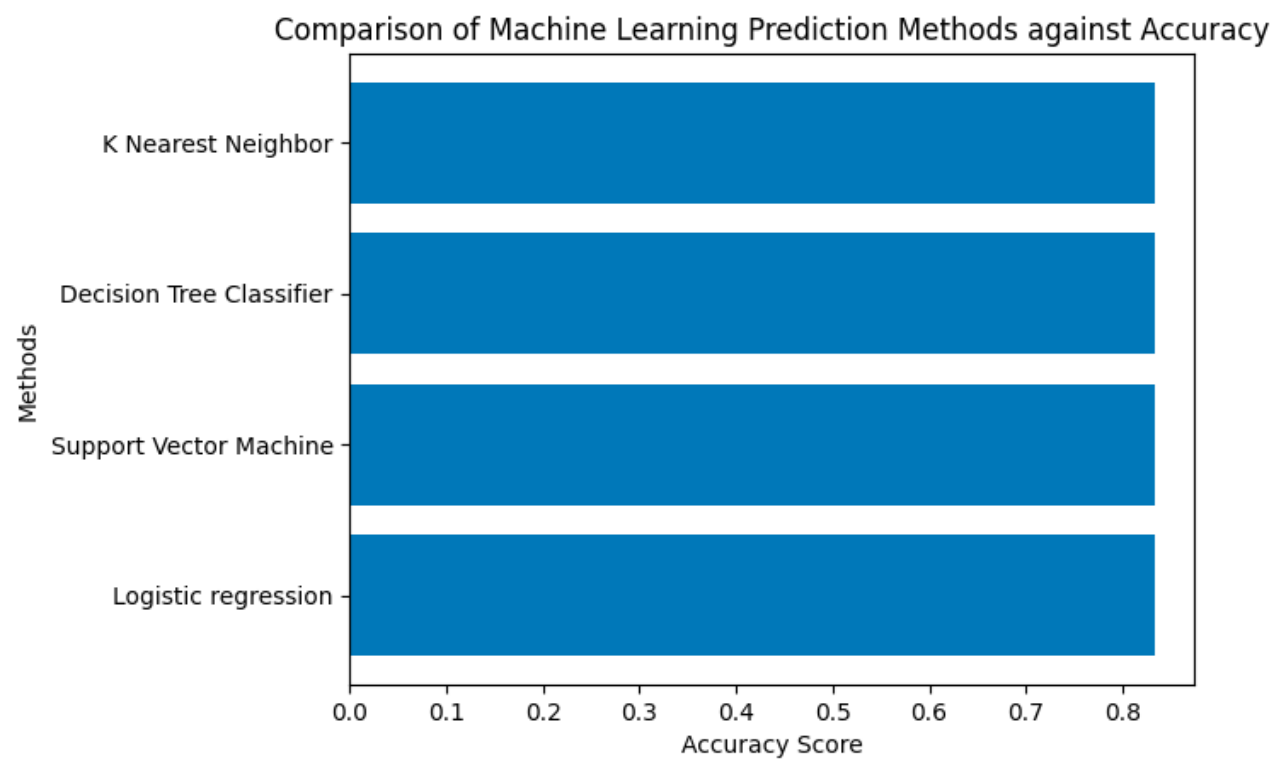
The launch success rate is higher when payload mass is less than 5,000 Kg.
FT is the most successful Booster Version.
Booster Version v1.1 has the highest number of failed launches.

Section 5

Predictive Analysis (Classification)

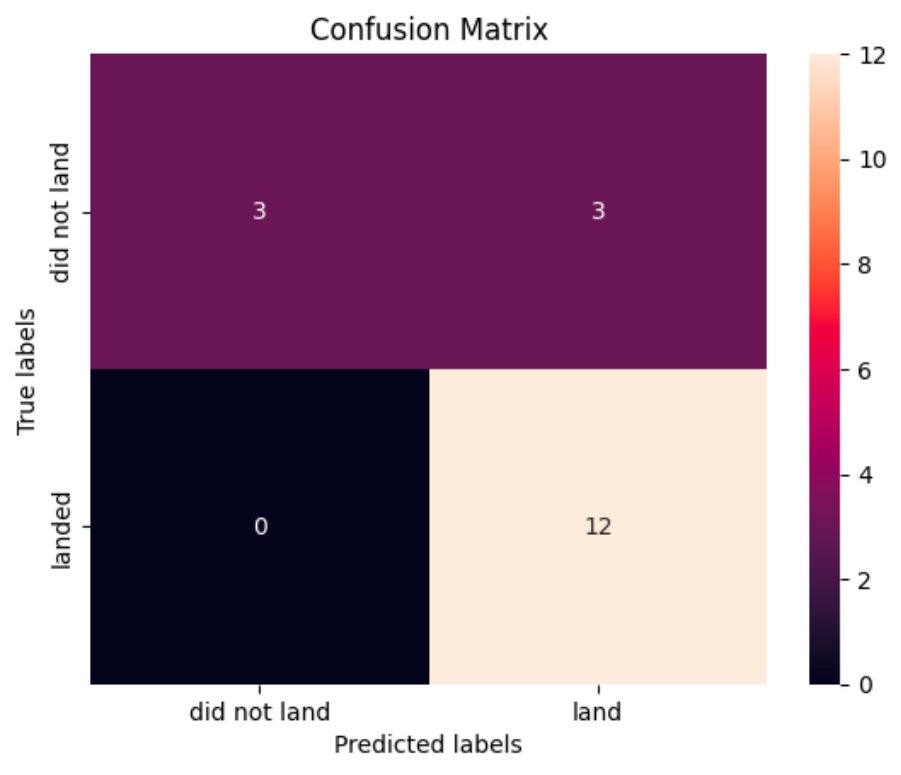


Classification Accuracy



Classification Accuracy for each Machine Learning Model is the same at 83.33%

Confusion Matrix



The confusion matrix for each model is the same.

Each model predicted 3 false positives and 0 false negatives.

Conclusions

- As the number of launches increases, the chance of a successful launch also increases
- Launches with lower payload mass have a higher success rate.
- Launches from the KSC LC-39A launch site have the highest success rate
- Launches to Orbits SSO, HEO, GEO and ES-L1 have a 100% success rate

We were able to build a machine language model that achieves 83.3% accuracy predicting launch success.

Appendix

- Coursera - IBM Data Science Professional Certificate Course
- GitHub - IBM-ADS-Capstone
- Open-Source SpaceX REST API
- Wikipedia's List of Falcon 9 and Falcon Heavy launches
- Folium
- Plotly Dash

Thank you!

